Status	Finished
Started	Friday, 16 October 2026, 8:01 PM
Completed	Friday, 16 October 2026, 8:04 PM
Duration	3 mins 5 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(log n) ✓
c. O(1)
d. O👎

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99
b. 90 and 99 ✓
c. 90 and 100
d. 89 and 94

Question **3**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Sequential search
c. Linear search
d. Binary search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is not the array at all

d. When the item is the last element in the array

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓

b. When the list is sorted

c. When the list contains duplicate elements

d. When the search algorithm finishes

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements

b. The search stops immediately ✓

c. The search starts over from the beginning

d. The search continues until the end of the list

Question **7**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Linear

c. Hash

d. Binary ✓

Question **8**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Selection

d. Insertion

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Must use a sorted array

c. There must be a mechanism to access middle element directly

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues from the beginning
c. The list is sorted
d. The search continues in the left or right sublist ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. An error is raised
c. The first element is returned
d. The last element is returned

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n ✓
d. n-1

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By dividing the list into halves

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 500
c. 10 ✓
d. 100

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Bubble search
c. Quick search
d. Linear search ✓